UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 12

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Now, let's move on from $\mathbf{A}\boldsymbol{x} = \boldsymbol{y}$ to LS:

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|_2,$$

where $\mathbf{A} \in \mathbb{R}^{m\times n}$, and $\boldsymbol{y} \in \mathbb{R}^m$ is the vector of measurements from which $\boldsymbol{x}$ is to be found.

- There is noise/error in $\boldsymbol{y}$.
- **Question:** How does changing $\boldsymbol{y}$ to $\boldsymbol{y} + \Delta\boldsymbol{y}$ affect solutions?
- Let $\mathcal{S}$ denote the set of solutions to $\min\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|_2$.
- We know that $\mathcal{S} = \mathbf{A}^+\boldsymbol{y} + \mathcal{N}(\mathbf{A})$.
- Let $\mathcal{S}_\text{p}$ denote the set of solutions to the perturbed LS:

  $$\min_{\boldsymbol{x}\in\mathbb{R}^n} \|\mathbf{A}\boldsymbol{x} - (\boldsymbol{y} + \Delta\boldsymbol{y})\|_2.$$

  Then $\mathcal{S}_\text{p} = \mathbf{A}^+(\boldsymbol{y} + \Delta\boldsymbol{y}) + \mathcal{N}(\mathbf{A})$.
- $\mathcal{S}$ and $\mathcal{S}_\text{p}$ are two parallel affine sets characterized by the same subspace $\mathcal{N}(\mathbf{A})$ but have different biased terms $\mathbf{A}^+\boldsymbol{y}$ and $\mathbf{A}^+(\boldsymbol{y} + \Delta\boldsymbol{y})$. Hence, the difference between $\mathcal{S}$ and $\mathcal{S}_\text{p}$ is $\mathbf{A}^+\Delta\boldsymbol{y}$.
- Let $\boldsymbol{x}^*$ and $\boldsymbol{x}^* + \Delta\boldsymbol{x}$ denote the minimum norm solutions to LS and the perturbed LS, then

  $$\mathcal{S} = \underbrace{\boldsymbol{x}^*}_{\mathbf{A}^+\boldsymbol{y}} + \mathcal{N}(\mathbf{A}), \quad \mathcal{S}_\text{p} = \underbrace{(\boldsymbol{x}^* + \Delta\boldsymbol{x})}_{\mathbf{A}^+(\boldsymbol{y}+\Delta\boldsymbol{y})} + \mathcal{N}(\mathbf{A}),$$

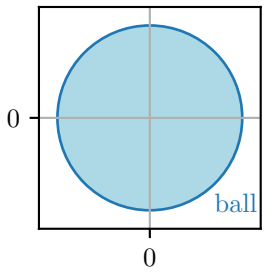
  and $\Delta\boldsymbol{x} = \mathbf{A}^+\Delta\boldsymbol{y}$.

Assume that $\Delta\boldsymbol{y}$ is an arbitrary perturbation with norm at most 1 (you can replace 1 with any small number): $\|\Delta\boldsymbol{y}\| \le 1$.
Define:

$$\mathcal{E} = \{\Delta\boldsymbol{x} = \mathbf{A}^+\Delta\boldsymbol{y} \mid \|\Delta\boldsymbol{y}\| \le 1\}.$$

Let's study sets of the form $\mathcal{E}$.

- First consider a ball centered at the origin of radius r in $\mathbb{R}^2$:



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- The ball can be expressed as

$$\begin{aligned}\text{ball} &= \{\boldsymbol{x} \in \mathbb{R}^2 \mid x_1^2 + x_2^2 \le r^2\} \\ &= \left\{\boldsymbol{x} \in \mathbb{R}^2 \;\middle|\; [x_1, x_2] \begin{bmatrix} \frac{1}{r^2} & 0 \\ 0 & \frac{1}{r^2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \le 1 \right\}.\end{aligned}$$

  Note that

$$[x_1, x_2] \begin{bmatrix} \frac{1}{r^2} & 0 \\ 0 & \frac{1}{r^2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = [x_1, x_2] \begin{bmatrix} \frac{1}{r} & 0 \\ 0 & \frac{1}{r} \end{bmatrix} \begin{bmatrix} \frac{1}{r} & 0 \\ 0 & \frac{1}{r} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left\| \begin{bmatrix} \frac{1}{r} & 0 \\ 0 & \frac{1}{r} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|^2,$$

  we have

$$\begin{aligned}\text{ball} &= \left\{\boldsymbol{x} \in \mathbb{R}^2 \;\middle|\; \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{r} & 0 \\ 0 & \frac{1}{r} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},\ \|\boldsymbol{y}\|^2 \le 1 \right\} \\ &= \left\{\boldsymbol{x} \in \mathbb{R}^2 \;\middle|\; \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{r} & 0 \\ 0 & \frac{1}{r} \end{bmatrix}^{-1} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix},\ \|\boldsymbol{y}\| \le 1 \right\} \\ &= \left\{\boldsymbol{x} \in \mathbb{R}^2 \;\middle|\; \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix},\ \|\boldsymbol{y}\| \le 1 \right\}.\end{aligned}$$

- Similarly, a ball in $\mathbb{R}^n$ can be expressed as

$$\begin{aligned}\text{ball} &= \{\boldsymbol{x} \in \mathbb{R}^n \mid x_1^2 + \cdots + x_n^2 \le r^2\} \\ &= \left\{\boldsymbol{x} \in \mathbb{R}^n \;\middle|\; \boldsymbol{x} = \begin{bmatrix} r & & 0 \\ & \ddots & \\ 0 & & r \end{bmatrix} \boldsymbol{y},\ \|\boldsymbol{y}\| \le 1 \right\}.\end{aligned}$$

- How about

$$\{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{x} = \mathbf{B}\boldsymbol{y},\ \|\boldsymbol{y}\| \le 1\}$$

  where $\mathbf{B} \in \mathbb{R}^{n \times n}$ is invertible? This is an *ellipse*:

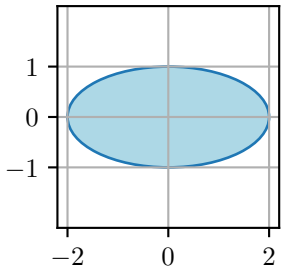
$$\begin{aligned}\{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{x} = \mathbf{B}\boldsymbol{y},\ \|\boldsymbol{y}\| \le 1\} &= \left\{\boldsymbol{x} \in \mathbb{R}^n \mid \|\mathbf{B}^{-1}\boldsymbol{x}\| \le 1\right\} \\ &= \left\{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{x}^{\mathrm{T}}\left(\mathbf{B}^{-1}\right)^{\mathrm{T}}\mathbf{B}^{-1}\boldsymbol{x} \le 1\right\} \\ &= \left\{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{x}^{\mathrm{T}}\mathbf{P}^{-1}\boldsymbol{x} \le 1\right\}, \qquad (1)\end{aligned}$$

  where $\mathbf{P} = \mathbf{B}\mathbf{B}^{\mathrm{T}}$ is positive definite.

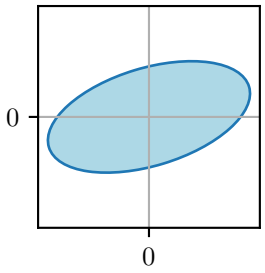
- Equation (1) is the definition of an n-dimensional ellipse.

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Example 1. Let $\mathbf{P} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$. Ellipse:

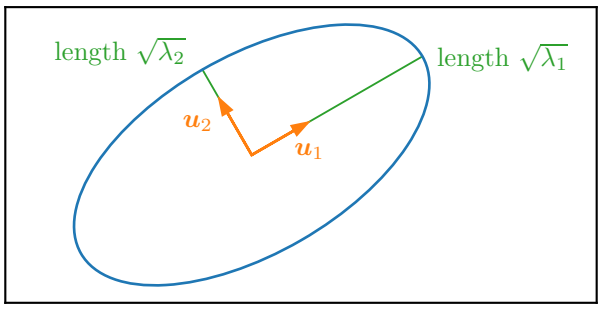


Let $\mathbf{P} = \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix}$. Ellipse:



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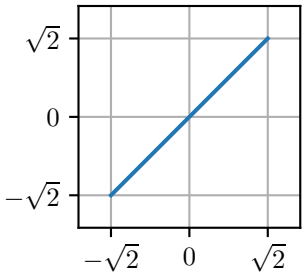
Let $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)}$ be the eigenvectors of $\mathbf{P}$ and $\lambda_1, \ldots, \lambda_n$ be the eigenvalues. Then, the ellipse has n semi-axes in the directions $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)}$ with lengths $\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}$.
Example in $\mathbb{R}^2$:



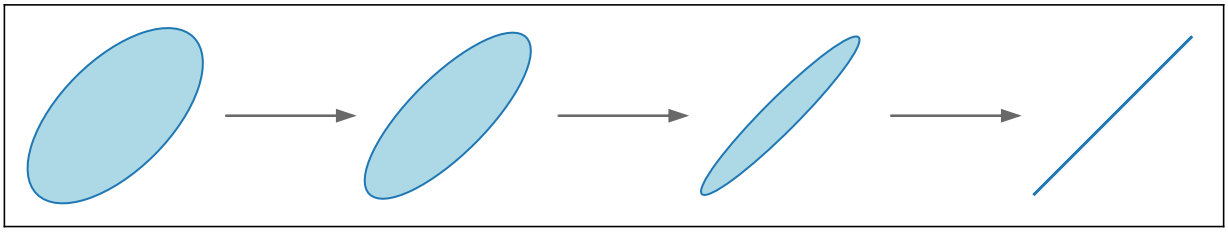
- Now consider the set
$$\{\boldsymbol{x} \mid \boldsymbol{x} = \mathbf{B}\boldsymbol{y},\ \|\boldsymbol{y}\| \leq 1\}$$
where $\boldsymbol{x} \in \mathbb{R}^n$ and $\boldsymbol{y} \in \mathbb{R}^m$. This makes $\mathbf{B} \in \mathbb{R}^{n \times m}$.
- $\mathbf{B}$ may not be square or invertible.

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Example 2. Let $n = m = 2$, $\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow \mathbf{B}\boldsymbol{y} = \begin{bmatrix} y_1 + y_2 \\ y_1 + y_2 \end{bmatrix}$. Then, the set $\{\boldsymbol{x} = \mathbf{B}\boldsymbol{y} \mid y_1^2 + y_2^2 \leq 1\}$ is a line segment with angle 45°:

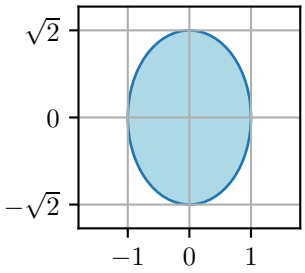


This is a degenerate (flat) ellipse:



(Make the ellipse narrower until we go from dimension 2 to dimension 1.) ■

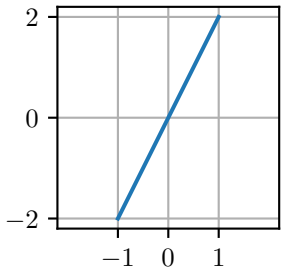
Example 3. Let $n = 2$, $m = 3$, $\mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \Rightarrow \mathbf{B}\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 + y_3 \end{bmatrix}$. Then, the set $\{\boldsymbol{x} = \mathbf{B}\boldsymbol{y} \mid y_1^2 + y_2^2 + y_3^2 \leq 1\}$ is an ellipse:



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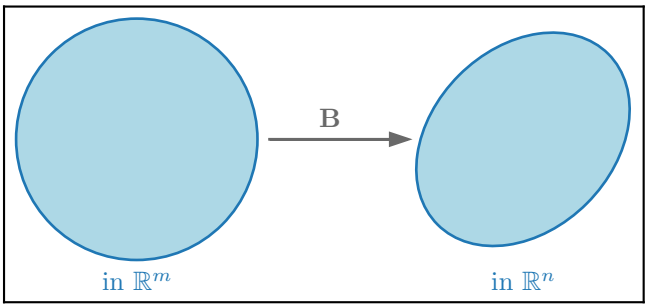
Example 4. Let $n = 2$, $m = 1$, $\boldsymbol{B} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \Rightarrow \boldsymbol{B}y = \begin{bmatrix} y \\ 2y \end{bmatrix}$. Then, the set $\{\boldsymbol{x} = \boldsymbol{B}y \mid y^2 \leq 1\}$ is a line segment:

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- The set $\{\boldsymbol{y} \in \mathbb{R}^m \mid \|\boldsymbol{y}\| \leq 1\}$ is a ball.
- The set $\{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{x} = \mathbf{B}\boldsymbol{y},\ \|\boldsymbol{y}\| \leq 1\}$ is a linear transformation of the ball by the matrix $\mathbf{B}$:

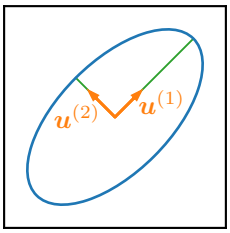


Theorem 1. *Consider the set* $\{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{x} = \mathbf{B}\boldsymbol{y},\ \|\boldsymbol{y}\| \leq 1\}$ *where* $\mathbf{B} \in \mathbb{R}^{n \times m}$.

- *Let* r *denote the rank of* $\mathbf{B}$ *and* $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$ *denote its non-zero singular values.*
- *Let* $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)}$ *be the left singular vectors of* $\mathbf{B}$.

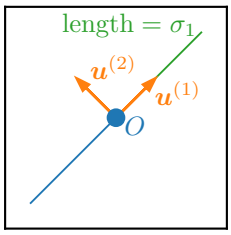
Then, the above set is an r*-dimensional ellipse with semi-axes* $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)}$ *and lengths* $\sigma_1, \ldots, \sigma_r, \underbrace{0, \ldots, 0}_{n-r}$.

Example 5 (Examples for $n = 2$)**.** If $\sigma_1, \sigma_2 > 0$:



If $\sigma_1 > 0, \sigma_2 = 0$:

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This is a degenerate ellipse, and the length along direction $\boldsymbol{u}^{(2)}$ is $\sigma_2 = 0$. ■

Going back to LS:

- $\mathcal{S}$: the set of solutions of $\min\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|$.

$$\mathcal{S} = \mathbf{A}^+\boldsymbol{y} + \mathcal{N}(\mathbf{A}).$$

- $\mathcal{S}_\text{p}$: the set of solutions of $\min\|\mathbf{A}\boldsymbol{x} - (\boldsymbol{y} + \Delta\boldsymbol{y})\|$.

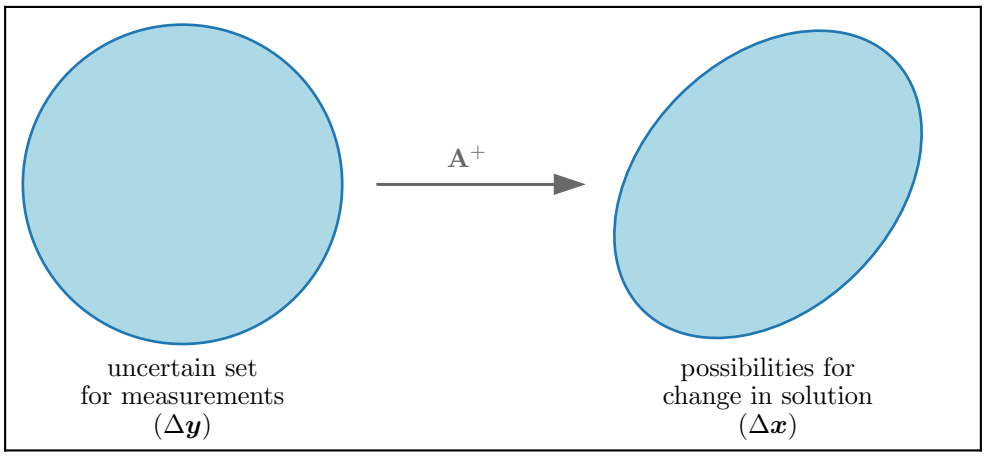
$$\mathcal{S}_\text{p} = \mathbf{A}^+(\boldsymbol{y} + \Delta\boldsymbol{y}) + \mathcal{N}(\mathbf{A}) = \mathcal{S} + \mathbf{A}^+\Delta\boldsymbol{y}.$$

- Optimal solutions: $\boldsymbol{x}^* = \mathbf{A}^+\boldsymbol{y}$, $\boldsymbol{x}^* + \Delta\boldsymbol{x} = \mathbf{A}^+(\boldsymbol{y} + \Delta\boldsymbol{y})$.
- Define

$$\mathcal{E} = \{\Delta\boldsymbol{x} \in \mathbb{R}^n \mid \Delta\boldsymbol{x} = \mathbf{A}^+\Delta\boldsymbol{y},\ \|\Delta\boldsymbol{y}\| \le 1\}.$$

  (Here 1 can be changed to any constant.)
- Note that $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\text{T} \Rightarrow \mathbf{A}^+ = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\text{T}$, where the left singular vectors of $\mathbf{A}^+$ are $\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(n)}$, and the singular values are $\frac{1}{\sigma_1}, \ldots, \frac{1}{\sigma_r}, 0, \ldots, 0$.
- By Theorem 1, $\mathcal{E}$ is an ellipse with semi-axes $\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(n)}$, and the lengths along these axes are $\frac{1}{\sigma_1}, \ldots, \frac{1}{\sigma_r}, \underbrace{0, \ldots, 0}_{n-r}$.

Example 6. Let $n = m = 2$. Since $\Delta\boldsymbol{x} = \mathbf{A}^+\Delta\boldsymbol{y}$:



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The set $\{\Delta \boldsymbol{y} \in \mathbb{R}^m \mid \|\Delta \boldsymbol{y}\| \leq 1\}$ on the left is a unit circle, while the set $\mathcal{E} = \{\Delta \boldsymbol{x} \in \mathbb{R}^n \mid \Delta \boldsymbol{x} = \mathbf{A}^+ \Delta \boldsymbol{y},\ \|\Delta \boldsymbol{y}\| \leq 1\}$ is an ellipse. ■

- For sensitivity of system of equations $\boldsymbol{y} = \mathbf{A}\boldsymbol{x}$, the condition number $\kappa(\mathbf{A}) = \frac{\sigma_1}{\sigma_n}$ was important.
- Ellipsoidal characterization for least-squares gives a better result since it involves *all* singular values $\sigma_1, \sigma_2, \ldots$.

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